

Learning Cubic Field Representation from A Single Panorama for Virtual Reality

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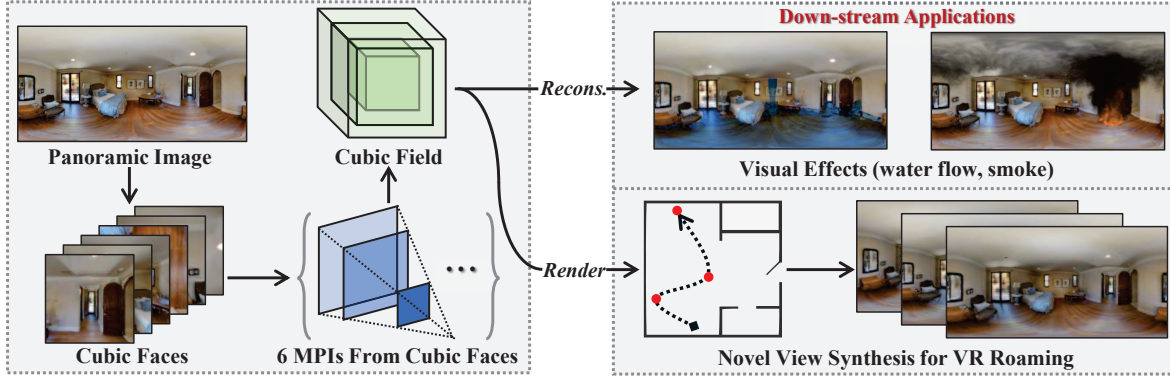


Figure 1: Our work constructs a cubic field representation of a scene captured by a 360-degree camera. It enables visual effects by providing depth maps and also supports VR roaming by rendering the scene from different viewpoints in real-time.

ABSTRACT

Panoramic images provide comprehensive scene information and are suitable for VR applications. Obtaining corresponding depth maps is essential for achieving immersive and interactive experiences. We propose a novel method that learns a cubic field composed of multiple MPIs from a single panoramic image for depth estimation. Our method employs cubemap projection to transform an ERP image into six faces and extract the MPIs for each. An attention-based blending module is employed to learn correlations among the MPIs, constructing a cubic field representation. Furthermore, a novel sampling strategy is introduced for rendering novel views from the cubic field at both cubic and planar scales. The entire pipeline is trained using photometric loss calculated from rendered views within a self-supervised learning approach. Experiments demonstrate the superior performance of the proposed method and highlight its effectiveness in downstream applications.

Index Terms: 360 Depth Estimation, Self-supervised Learning, Volume Rendering, Multi-Plane Images.

1 INTRODUCTION

Since panoramic depth information can greatly enhance the realism and interactivity of virtual environments by accurately mapping the 3D geometry of the surrounding scene, the ability to infer depth from a single 360-degree image has driven a large suite of research endeavors for monocular 360 depth estimation. Existing work primarily adopts a supervised approach: These methods directly generate depth maps from individual panoramic images by training on RGB-D datasets. Several recent works have explored self-supervised panoramic depth estimation, which trains the depth estimation network by rendering images at different viewpoints and constructing photometric loss. As depth maps fail to capture the

content hidden in the reference view but revealed in the target view, rendering novel views from depth maps is insufficient. To overcome this limitation, researchers adopt MPI representation [4] that models 3D space with a set of front-parallel layers to generate satisfying renderings under disocclusions and non-Lambertian effects and thus produce reasonable depth maps. However, the high resolution and inherent distortions of panoramic images pose significant challenges for MPI-based 3D representations. Cubemap projection addresses these issues by dividing the 360-degree content into six separate 2D images, each corresponding to one face of a cube. This approach not only reduces distortion but also lowers the resolution of each image, as demonstrated in prior work [5]. Building on these benefits, we introduce a novel panoramic representation based on cube-wise MPI, termed the cubic field. In our proposed pipeline, one panoramic image is first divided into six faces of a cubemap. An encoder-decoder based network takes these cubic faces as inputs and predicts the related MPIs that reconstruct the color and density information of a conical space at the pre-defined depths separately. Subsequently, the independent predicted MPIs of six faces are fed into a series of blending modules to generate the cubic field. A dual sampling strategy combined with volume rendering techniques is adopted to synthesize novel views from the cubic field at both the cubic and planar scales, which are further adopted to construct the photometric losses for supervision. We evaluate our method on synthetic and real-world datasets and show that it outperforms state-of-the-art methods.

2 METHODOLOGY

Cubic Field. Given a panoramic image, we first project it to six faces $\{\mathbf{f}_i \in \mathbb{R}^{w \times w} | i \in [B, D, F, L, R, U]\}$ of a cubemap, which is denoted as $\mathbf{f}_i = E2C(\mathbf{p})$, where $\mathbf{p} \in \mathbb{R}^{H \times W}$ is the input panoramic image. **MPI**ⁱ are predicted by an encoder-decoder network from $\mathbf{f}_i$. An **MPI**ⁱ includes the radiance $c_z \in \mathbb{R}^{w \times w \times 3}$ and density $\sigma_z \in \mathbb{R}^{w \times w \times 1}$ of d planes, where z denotes the pre-defined depth value of the related plane. We denote the set of d depth values as $D = \{z_1, z_2, \dots, z_d\}$. Then, **MPI**ⁱ is tokenized and processed through a series of modules, including self-attention and cross-attention from the transformer architecture [1], as well as padding operation introduced in [2]. These steps collectively produce the

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Table 1: Experimental comparisons on different datasets.

Dataset	Ours	Bifuse++ [5]	MINE [4]
PanoSUNCG	0.3839	0.4496	0.4190
Matterport3D	0.3349	0.5173	0.4079
Stanford2D3D	0.3209	0.5134	0.4218

final cubic field representation.

Novel View Rendering To achieve novel view rendering, which is subsequently utilized in the construction of photometric loss, we first integrate the standard inverse homography with the process of generating cubemap images. This integration allows us to define the Planar Homography Sampling method, as referenced in [3]. Based on this, we establish the mapping $c_z(m_t, n_t) = c_z(m_s, n_s)$ and $\sigma_z(m_t, n_t) = \sigma_z(m_t, n_t)$. Using this mapping, the synthesized target cubic face $\hat{\mathbf{f}}$ is rendered from the volume rendering scheme introduced in [4]. Next, we utilize Ray-Cube Sampling to establish a relationship between a unit sphere corresponding to the target view and the constructed cubic field. This process enables us to obtain the synthesized target panoramic image $\hat{\mathbf{I}}_{gr}$.

Loss Function The photometric loss minimizes the difference between the target image and the rendered image, which is formulated as:

$$\mathcal{L} = \frac{1}{3HW} \sum |\hat{\mathbf{I}} - \mathbf{I}| + \frac{1}{6w^2} \sum |\hat{\mathbf{f}} - E2C(\mathbf{I})| + 1 - \text{SSIM}(\hat{\mathbf{I}}, \mathbf{I}) + 1 - \text{SSIM}(\hat{\mathbf{f}}, E2C(\mathbf{I})),$$

where $\mathbf{I}$ is the target panorama, $E2C(\mathbf{I})$ are target cubic faces, $\hat{\mathbf{I}}$ is the rendered panorama and $\hat{\mathbf{f}}$ are rendered cubic faces.

3 RESULTS

Quantitative Comparisons. To evaluate our method, we conduct comparisons against the state-of-the-art (SOTA) self-supervised methods, BiFuse++ [5] and MINE [4], as shown in Table 1. We assess the performance by measuring the RMSE between the predicted depth maps and the ground truth. The results show that the proposed framework achieves the best performance.

VR Wondering. By leveraging this cubic representation, panoramic images from various viewpoints are generated via volume rendering techniques. This approach allows for scene navigation, enabling users to virtually "roam" through the reconstructed environment. As shown in Fig. 2, novel views are rendered along a predefined trajectory, showcasing the scene from different positions. Our method achieves a real-time rendering speed of 414 fps on an NVIDIA RTX A6000.

Real-World Adaptation and Evaluation. Our method is built on self-supervised learning, allowing it to be fine-tuned on videos captured by consumer-level panoramic cameras without the need for ground truth depth information. This flexibility enhances its practicality and efficiency in real-world applications. We conducted experiments on self-captured real-world videos and compared the results to those produced by supervised methods. As shown in Fig. 3, our approach outperforms supervised methods, particularly in challenging regions, further demonstrating its effectiveness in practical scenarios. These findings highlight the potential of our method as a more robust and efficient solution for depth inference from consumer-grade panoramic cameras.

4 CONCLUSION AND LIMITATIONS

We propose a novel framework to learn cubic field representation from a single panorama using a self-supervised approach, which enables 360 depth estimation and novel view synthesis. Our method learns a cubic field representation to model the color and density

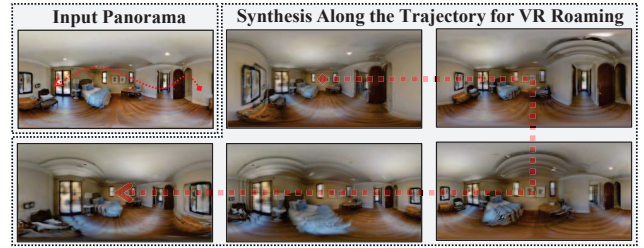


Figure 2: Input panorama and rendered novel views along a predefined trajectory. The top-left shows the original input panorama, while the remaining images present novel views generated along the indicated path (red dotted lines). These novel views highlight the effectiveness of our method in enabling immersive VR experiences with realistic scene exploration.

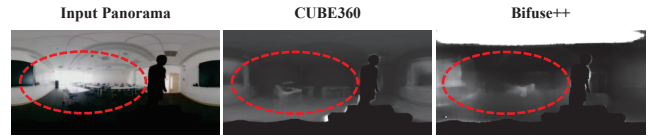


Figure 3: Comparison of depth inference results on real-world panoramic images captured with RICOH THETA Z1 camera. The input panorama (left) is processed using the proposed method (middle), demonstrating enhanced depth prediction quality, especially in challenging regions (highlighted in red), compared to traditional supervised methods (right).

information of an entire scene. We introduce a dual-sampling strategy that facilitates novel view synthesis at both cubic and planar scales within the cubic field. Additionally, we develop an attention-based blending model to integrate cross-face features for generating the cubic field. Experimental results demonstrate that the proposed method outperforms state-of-the-art (SOTA) methods on both synthetic and real-world datasets. However, due to limitations in the dataset and the size of current neural network models, our method is unable to process high-resolution image data up to 4K. Addressing this challenge remains a key direction for future research.

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